

BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination: Semester Final
Duration: 1 hour 45 min

Semester: Spring 2025
Full Marks: 45

CSE 320/EEE361/ECE361: Data Communications

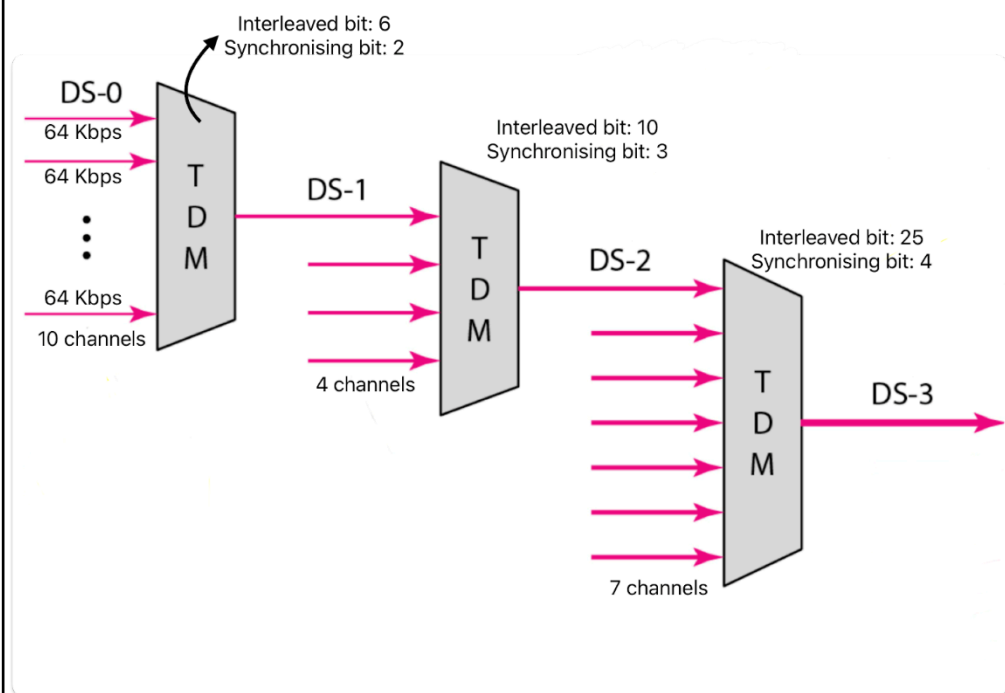
Answer the following questions.
Figures in the right margin indicate marks.

SET A

Name:	ID:	Section:
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1. [CO2]	a) In a QPSK, the amplitudes and the phases are as follows for pairs of bits. <table><tr><td>Bits</td><td>Amplitude</td><td>Phase</td></tr><tr><td>00</td><td>5</td><td>45</td></tr><tr><td>01</td><td>5</td><td>135</td></tr><tr><td>10</td><td>5</td><td>315</td></tr><tr><td>11</td><td>5</td><td>225</td></tr></table> The carrier frequency contains 2.5 cycles for each signal element. Given the bit sequence 1011000110, Draw the analog signal. Amplitude 5V 0V -5V Time	Bits	Amplitude	Phase	00	5	45	01	5	135	10	5	315	11	5	225	[5]
Bits	Amplitude	Phase															
00	5	45															
01	5	135															
10	5	315															
11	5	225															
2. [CO3]	a) A control system has 8 devices. In the first time slot, only devices 1, 3, 5, 6, and 8 have data. In the second time slot, devices 1, 2, 4, 5, and 7 have data. • What TDM technique should be used here? • Sketch a figure of the TDM process and show the frames. • If three bits are used for addressing, determine the frame size. • Does this scheme have any drawbacks? Justify.	[1+2+1+1]															

b) In this hierarchical TDM, **calculate** the frame rate and output rate for DS-3. [7]



c) In a FDM, 4 input channels are multiplexed together. The size of the assigned bandwidth for each channel are 5KHz, 10 KHz, 15 KHz and 20 KHz respectively. If the guard band is 3KHz. What is the overall **required bandwidth** of the output channel? [3]

3.
[CO4]

a) Suppose a sender is sending a message consisting of five 5-bit numbers to the receiver. In both ends, traditional checksum is being used to detect errors. [4]

Sender		Receiver	
Data	8 12 14 0 9	Data	9 11 14 0 9
Actual Checksum	?	Received Checksum	19

- Is the actual checksum from the sender equal to the checksum received by the receiver?
- Are there any errors in the data received? **Explain** if the errors are detectable using checksum or not.

b)	Observe the Block Coding scheme below. <table><tr><th>Dataword</th><th>Codeword</th></tr><tr><td>00</td><td>0001110</td></tr><tr><td>01</td><td>0110110</td></tr><tr><td>10</td><td>1010101</td></tr><tr><td>11</td><td>1100001</td></tr></table> • Calculate the minimum hamming distance. • How many errors can this scheme detect and how many can it correct? • Receiver receives the two codewords 0110010 101110. Identify if there were any errors in the received codewords. For each codeword, find the original data after error correction?	Dataword	Codeword	00	0001110	01	0110110	10	1010101	11	1100001	[6]
Dataword	Codeword											
00	0001110											
01	0110110											
10	1010101											
11	1100001											
c)	Suppose you want to crack promo codes for your favorite “BAAPJI” games. You know that the game publishers are using CRC to verify the legitimacy of an entered code. Upon some research, you partially cracked the following info for one of your entries. • Codeword: $x^6 + x^5 + x$ • Quotient: $x^3 + x^2 + x$ • $C(7,4)$ However, sigh! No mention of the divisor. But a genius like you would never stop to crack your favorite game! So, you combined your current Data Comm. lesson and your high school math formula – $Dividend = Divisor * Quotient \oplus Remainder$ Now show us how you did it by answering the following questions. - From the given Codeword and $C(n,k)$, find out what the remainder was? Why so? - What was the dividend? Why so? - From (i), (ii), and the formula, find out the correct divisor. Show full steps. <i>Hint: Binary and Polynomial CRC are interchangeable. You may use any of them, whichever feels suitable for a question. All operations are in modulo-2 arithmetic.</i>	[1+1+3]										

4. [CO4]	a) Four devices are sharing a single communication channel using the CSMA/CD protocol as shown in the figure below. Propagation speed of the medium is 2×10^6 m/s. <pre> graph LR A[A] --- 2.4 m B[B] B --- 3.6 m C[C] C --- 2.0 m D[D] </pre> • Suppose, device A wants to send a frame to another device. If the channel bandwidth is 2 Mbps (megabits per sec), what will be the minimum frame size required? What if the frame size is smaller than that? • Suppose, device B starts sending a frame at time $2.0 \mu\text{s}$ (microsecond). Device D starts sending a frame at $4.5 \mu\text{s}$. Find out whether this will lead to a collision.	[3+3]
	b) Suppose, you are developing a remote sensor system with one device transmitting data infrequently over a communication channel. Since it is a remote setup, you need to minimize any overhead as much as possible. Which multiple access protocol should be used here? Explain.	[2]
	c) Three devices are sharing a common channel using pure ALOHA protocol. Frame transmission time is $10 \mu\text{s}$ (each frame takes $10 \mu\text{s}$ to transmit). • Device A generates TWO frames at times $10 \mu\text{s}$ and $40 \mu\text{s}$. • Device B generates TWO frames at times $20 \mu\text{s}$ and $35 \mu\text{s}$. • Device C generates ONE frame at time $5 \mu\text{s}$. Using a figure, find out how many frames will be successfully transmitted.	[2]

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